

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES	
10							***** MODULAR VALVES *****			B1	ALL BOLTS ARE METRIC DIA.
20										B2	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AM10
20	U04		MODULAR DBB	K03AC02	0.5	2	MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV	0	0		
30	U04	VAV	MODULAR DBB	K03AC02	3	3	MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV	0	0	B6	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AS22
40	U04		MODULAR DBB	M03AC02	0.5	2	MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR	0	0	SB6	USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS.
50	U04	VAV	MODULAR DBB	M03EC02	0.75	1	MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 CL.1 316SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	2	2	VA5	USE THESE VALVES ONLY WHERE SPECIFIED ON PIPING AND INSTRUMENT DIAGRAMS
60	U04	VAV	MODULAR DBB	M03EC02	1	1	MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 CL.1 316SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	3	3	VAV	USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE.
70							***** BALL VALVES *****			VAW	VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS.
70		VB2								VB2	SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.
80										VC1	SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL.
80	U04		BALL	A03AC01	3	4	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT RB LO	0	0		
90	U04		BALL	A03AC02	3	4	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING METAL RB LO	0	0		
100	U04		BALL	A03AC07	6	8	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT RB GO	0	0		
110	U04		BALL	A03AC08	6	8	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION METAL RB GO	0	0		
120	U04		BALL	B03AC01	0.75	2	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT FB LO	0	0		
130	U04	VA5	BALL	B03AC01	3	3	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT FB LO	0	0		
140	U04		BALL	B03AC02	0.75	2	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING METAL FB LO	0	0		
150	U04	VA5	BALL	B03AC02	3	3	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING METAL FB LO	0	0		
160	U04	VA5	BALL	B03AC03	4	4	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT FB GO	0	0		
170	U04	VA5	BALL	B03AC04	4	4	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING METAL FB GO	0	0		
180	U04	VA5	BALL	B03AC07	6	24	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT FB GO	0	0		
190	U04	VA5	BALL	B03AC08	6	24	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION METAL FB GO	0	0		
200	U04	VA5	BALL	B03JC07	28	28	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT FB GO B16.47 SERIES A	0	0		
210	U04	VA5	BALL	B03JC08	28	28	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION METAL FB GO B16.47 SERIES A	0	0		
220							***** RISING STEM BALL VALVES *****				
220											
230	U04	VA5	BALL	R03AC06	2	8	BALL RISING STEM 300# RF LTCS A352 LCC 316SS TRIM METAL RB HO	0	0		
240	U04	VA5	BALL	R03AC08	10	10	BALL RISING STEM 300# RF LTCS A352 LCC 316SS TRIM METAL RB GO	0	0		
250							***** BUTTERFLY VALVES *****				
250											
260	U04		BUTTERFLY	T03AC61	10	24	BUTTERFLY 300# RF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO	0	0		
270			BUTTERFLY	T03JC61	28	28	BUTTERFLY 300# RF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO ASME B16.47 SERIES A	0	0		
280							***** GATE VALVES *****				
280											
290	U04		GATE	G03AC48	0.75	1.5	GATE 300# RF LTCS A350 LF2 CL.1 TRIM 16 WEDGE HW	0	0		
290			GATE	G03AC41	2	2	GATE 300# RF LTCS A352 LCC TRIM 16 WEDGE HW	0	0		
300			GATE	W03AC70	3	6	GATE 300# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB HW	0	0		
310	U04		GATE	W03AC71	8	20	GATE 300# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO	0	0		
320	U04		GATE	W03AC71	8	20	GATE 300# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO	0	0		
330							***** GLOBE VALVES *****				
330											
340	U04		GLOBE	L03AC48	0.75	1.5	GLOBE 300# RF LTCS A350 LF2 CL.1 TRIM 16 OS&Y HW	0	0		
340			GLOBE	L03AC41	2	6	GLOBE 300# RF LTCS A352 LCC TRIM 16 OS&Y HW	0	0		
350			GLOBE	L03AC42	8	10	GLOBE 300# RF LTCS A352 LCC TRIM 16 OS&Y GO	0	0		
360							***** CHECK VALVES *****				
370											
370											
380	U04	VC1	CHECK	C03AC50	0.75	1.5	CHECK PISTON 300# RF LTCS A350 LF2 CL.1 TRIM 16	0	0		
380			CHECK	C03AC45	2	10	CHECK DUAL WAFER LUG 300# RF LTCS A352 LCC TRIM 16 THROU BOLTED	0	0		
390			CHECK	C03AC52	12	24	CHECK DUAL DBL FLG 300# RF LTCS A352 LCC TRIM 16	0	0		
400			CHECK	C03JC52	28	28	CHECK DUAL DBL FLG 300# RF LTCS A352 LCC TRIM 16 B16.47 SERIES A	0	0		
410							***** PIPE ABOVE GROUND *****				
420											
420											
430			PIPE	5358508	0.75	1.5	PIPE SCH XS SMLS LTCS A333 GR.6 PE	0	0		
430			PIPE	5358514	2	2	PIPE SCH XS SMLS LTCS A333 GR.6 BE	0	0		
440			PIPE	5358505	3	3	PIPE SCH STD SMLS LTCS A333 GR.6 BE	0	0		
450											
460			PIPE	9001257	4	4	LINEPIPE SMLS WT= 6.02 MM API-5L X60 BE	0	0		
460			PIPE	9001254	6	10	LINEPIPE SMLS WT= 6.35MM API-5L X60 BE	0	0		
470			PIPE	9001499	12	12	LINEPIPE SMLS WT = 7.09 MM API-5L X60 BE	0	0		
480			PIPE	9001311	14	14	LINEPIPE SMLS WT=7.92MM API-5L X60 BE	0	0		
490			PIPE	9001485	16	18	LINEPIPE SMLS WT = 8.74 MM API-5L X60 BE	0	0		
500			PIPE	9001485	16	18	LINEPIPE SMLS WT = 8.74 MM API-5L X60 BE	0	0		

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD SEAL WELDING - REQUIRED PIPE BENDS - INDUCTION BENDS THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 3 MM	SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 47 BARG. @ 90 DEG.C. (LIMITED BY PIPE WT) FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 300 RF TEMPERATURE RANGE: -40 / 90 °C	12/16/2024	BY OZAC	U04	Page 1 of 4	300H5F

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES	
510			PIPE	9001493	20	20	LINEPIPE SMLS WT = 9.53 MM API-5L X60 BE	0	0		
520							***** PIPE BELOW GROUND *****				
530			PIPE	9001257CTD	4	4	LINEPIPE SMLS WT= 6.02 MM API-5L X60 BE COATED	0	0		
540			PIPE	9001254CTD	6	10	LINEPIPE SMLS WT= 6.35 MM API-5L X60 BE COATED	0	0		
550			PIPE	9001499CTD	12	12	LINEPIPE SMLS WT= 7.09 MM API-5L X60 BE COATED	0	0		
560			PIPE	9001311CTD	14	14	LINEPIPE SMLS WT= 7.92 MM API-5L X60 BE COATED	0	0		
570			PIPE	9001485CTD	16	18	LINEPIPE SMLS WT = 8.74 MM API-5L X60 BE COATED	0	0		
580			PIPE	9001493CTD	20	20	LINEPIPE SMLS WT = 9.53 MM API-5L X60 BE COATED	0	0		
590							***** FITTINGS ABOVE & BELOW GROUND*****				
600			ELBOW	5392600	0.75	2	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6	0	0		
610			ELBOW	5392438	3	3	ELL BW 90 DEG LR SCH STD LTCS A420 WPL6	0	0		
620			ELBOW	5392606	0.75	2	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6	0	0		
630			ELBOW	5392444	3	3	ELL BW 45 DEG LR SCH STD LTCS A420 WPL6	0	0		
640											
640			TEE	5392609	0.75	2	TEE BW SCH XS LTCS A420 WPL6	0	0		
650			TEE	5392447	3	3	TEE BW SCH STD LTCS A420 WPL6	0	0		
660			TEE RED	5392661	0.75	1.5	TEE RED BW SCH XS LTCS A420 WPL6	1	2		
670			TEE RED	5392454	0.75	2	TEE RED BW SCH STD/XS LTCS A420 WPL6	3	3		
680			TEE RED	9006200	4	4	TEE RED BW WT 7.09 MM / 6.02MM LTCS A860 WPHY 60	12	12		
690			TEE RED	9004123	6	6	TEE RED BW WT 8.74 MM/6.35 MM LTCS A860 WPHY 60	16	16		
700											
700			CAP	5392610	0.75	2	CAP BW SCH XS LTCS A420 WPL6	0	0		
710			CAP	5392448	3	3	CAP BW SCH STD LTCS A420 WPL6	0	0		
720			CAP	9005055	6	10	CAP BW WT 6.35MM LTCS A860 WPHY-60	0	0		
730			CAP	9005063	12	12	CAP BW WT 7.09 MM LTCS A860 WPHY-60	0	0		
740			CAP	9005058	14	14	CAP BW WT 7.92 MM LTCS A860 WPHY-60	0	0		
750			CAP	9005059	16	18	CAP BW WT 8.74 MM LTCS A860 WPHY-60	0	0		
760			CAP	9005060	20	20	CAP BW WT 9.53 MM LTCS A860 WPHY-60	0	0		
770											
770			RED CONC	5392458	0.75	1.5	REDUCER CONC SCH XS LTCS A420 WPL6	1	2		
780			RED ECC	5392461	0.75	1.5	REDUCER ECC SCH XS LTCS A420 WPL6	1	2		
790			RED CONC	5392452	1.5	2	REDUCER CONC SCH STD/XS LTCS A420 WPL6	3	3		
800			RED ECC	5392453	1.5	2	REDUCER ECC SCH STD/XS LTCS A420 WPL6	3	3		
810			RED ECC	9002281	3	3	REDUCER ECC WT 6.02 MM / STD WT LTCS A860 WPHY-60	4	4		
820			RED ECC	9006103	3	3	REDUCER ECC WT 6.35 MM / STD LTCS A860 WPHY-60	6	8		
830			RED ECC	9002312	4	4	REDUCER ECC WT 6.35 MM / 6.02 MM LTCS A860 WPHY-60	6	10		
840			RED ECC	9002266	6	8	REDUCER ECC WT 6.35 MM / 6.35 MM LTCS A860 WPHY-60	8	10		
850			RED ECC	9006000	4	4	REDUCER ECC WT 7.09MM / 6.02 MM LTCS A860 WPHY-60	12	12		
860			RED ECC	9002416	6	10	REDUCER ECC WT 7.09 MM / 6.35 MM A860 WPHY 60	12	12		
870			RED ECC	9002272	6	10	REDUCER ECC WT 7.92 MM / 6.35 MM LTCS A860 WPHY-60	14	14		
880			RED ECC	9002292	8	10	REDUCER ECC WT 8.74 MM / 6.35 MM LTCS A860 WPHY-60	16	16		
890			RED ECC	9002250	14	14	REDUCER ECC WT 8.74 MM / 7.92 MM LTCS A860 WPHY-60	16	18		
900			RED ECC	9004135	14	14	REDUCER ECC WT 9.53 MM / 7.92MM LTCS A860 WPHY-60-W	20	20		
910			RED ECC	9004136	16	18	REDUCER ECC WT 9.53 MM / 8.74 LTCS A860 WPHY-60-W	20	20		
920							*****INDUCTION BENDS ABOVE GROUND*****				
930											
930			INDUC BEND	9001810	4	4	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0		
940			INDUC BEND	9001813	6	10	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.35MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0		
950			INDUC BEND	9005430	12	12	INDUCTION BEND 90 DEG X 5D RADIUS WT 7.09 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0		
960			INDUC BEND	9001815	14	14	INDUCTION BEND 90 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0		
970			INDUC BEND	9001816	16	18	INDUCTION BEND 90 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0		
980			INDUC BEND	9001817	20	20	INDUCTION BEND 90 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0		
990	U04		INDUC BEND	9001820	4	4	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0		
1000			INDUC BEND	9001820	4	4	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0		
1010			INDUC BEND	9001823	6	10	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0		
1020			INDUC BEND	9005431	12	12	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.09 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0		
1030			INDUC BEND	9001825	14	14	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0		
1040			INDUC BEND	9001826	16	18	INDUCTION BEND 45 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0		
1050			INDUC BEND	9001827	20	20	INDUCTION BEND 45 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0		
1060			INDUC BEND	9001830	4	4	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0		
1070			INDUC BEND	9001833	6	10	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0		
1080			INDUC BEND	9005432	12	12	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.09 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0		
1090			INDUC BEND	9001835	14	14	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0		
1100			INDUC BEND	9001836	16	18	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0		
1110			INDUC BEND	9001837	20	20	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0		
1120							*****INDUCTION BENDS BELOW GROUND*****				
1120											

	TENGIZCHEVROI TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD SEAL WELDING - REQUIRED PIPE BENDS - INDUCTION BENDS THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 3 MM	SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 47 BARG. @ 90 DEG.C. (LIMITED BY PIPE WT) FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 300 RF TEMPERATURE RANGE: -40 / 90 °C	12/16/2024	BY OZAC	U04	Page 2 of 4	300H5F

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES	
1130											
1130			INDUC BEND	9001810CTD	4	4	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0		
1140			INDUC BEND	9001813CTD	6	10	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.35MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0		
1150			INDUC BEND	9005430CTD	12	12	INDUCTION BEND 90 DEG X 5D RADIUS WT 7.09 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0		
1160			INDUC BEND	9001815CTD	14	14	INDUCTION BEND 90 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0		
1170			INDUC BEND	9001816CTD	16	18	INDUCTION BEND 90 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0		
1180	U04		INDUC BEND	9001817CTD	20	20	INDUCTION BEND 90 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0		
1190			INDUC BEND	9001820CTD	4	4	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0		
1200			INDUC BEND	9001823CTD	6	10	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0		
1210	U04		INDUC BEND	9007001CTD	10	10	INDUCTION BEND 45DEG X 10D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0		
1220	U04		INDUC BEND	9007016CTD	10	10	INDUCTION BEND 45DEG X 15D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0		
1230	U04		INDUC BEND	9007031CTD	10	10	INDUCTION BEND 90 DEG X 10D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0		
1240	U04		INDUC BEND	9007043CTD	10	10	INDUCTION BEND 90 DEG X 15D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0		
1250			INDUC BEND	9005431CTD	12	12	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.09 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0		
1260			INDUC BEND	9001825CTD	14	14	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0		
1270			INDUC BEND	9001826CTD	16	18	INDUCTION BEND 45 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0		
1280			INDUC BEND	9001827CTD	20	20	INDUCTION BEND 45 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0		
1290			INDUC BEND	9001830CTD	4	4	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0		
1300			INDUC BEND	9001833CTD	6	10	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0		
1310			INDUC BEND	9005432CTD	12	12	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.09 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0		
1320			INDUC BEND	9001835CTD	14	14	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0		
1330			INDUC BEND	9001836CTD	16	18	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0		
1340			INDUC BEND	9001837CTD	20	20	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0		
1350							***** OLETS ABOVE & BELOW GROUND*****				
1350											
1360											
1360			OLETS	9006493	0.75	2	WELDOLET WT 6.02MM/SCH XS LTCS A694 F60	4	4		
1370			OLETS	9006494	3	3	WELDOLET WT 6.02MM/SCH STD LTCS A694 F60	4	4		
1380			OLETS	9006407	0.75	2	WELDOLET WT 6.35 MM/XS WT LTCS A694 F60	6	10		
1390			OLETS	9006409	3	3	WELDOLET WT 6.35 MM/STD WT LTCS A694 F60	6	10		
1400			OLETS	9006490	0.75	2	WELDOLET WT 7.09 MM/XS WT LTCS A694 F60	12	12		
1410			OLETS	9006491	3	3	WELDOLET WT 7.09MM/STD WT LTCS A694 F60	12	12		
1420			OLETS	9002552	0.75	2	WELDOLET WT 7.92MM/ XS WT LTCS A694 F60	14	14		
1430			OLETS	9006415	3	3	WELDOLET WT 7.92 MM/STD WT LTCS A694 F60	14	14		
1440			OLETS	9006414	4	4	WELDOLET WT 7.92 MM/6.02 MM LTCS A694 F60	14	14		
1450			OLETS	9006423	0.75	2	WELDOLET WT 8.74 MM/XS WT LTCS A694 F60	16	18		
1460			OLETS	9006422	3	3	WELDOLET WT 8.74 MM/STD WT LTCS A694 F60	16	18		
1470			OLETS	9006421	4	4	WELDOLET WT 8.74 MM/6.02 MM LTCS A694 F60	16	18		
1480			OLETS	9002008	6	10	WELDOLET WT 8.74 MM/6.35 MM LTCS A694 F60	16	18		
1490											
1490	U04		FLANGE	5504021	0.75	2	NIPOFLANGE 300# RF SCH STD/XS LTCS A350 LF2 CL.1	3	3		
1500			FLANGE	9002639	0.75	2	NIPOFLANGE 300# RF 6.02 MM/XS WT LTCS A694 F60	4	4		
1510			FLANGE	9002643	0.75	2	NIPOFLANGE 300# RF 6.35 MM/XS WT LTCS A694 F60	6	10		
1520			FLANGE	9006492	0.75	2	NIPOFLANGE 300# RF 7.09 MM/XS WT LTCS A694 F60	12	12		
1530			FLANGE	9002645	0.75	2	NIPOFLANGE 300# RF 7.92 MM/XS WT LTCS A694 F60	14	14		
1540			FLANGE	9002646	0.75	2	NIPOFLANGE 300# RF 8.74 MM/XS WT LTCS A694 F60	16	18		
1550			FLANGE	9002647	0.75	2	NIPOFLANGE 300# RF 9.53 MM/XS WT LTCS A694 F60	20	20		
1560											
1560							***** FLANGES ABOVE & BELOW GROUND*****				
1570											
1570	U04		FLANGE	5556476	0.75	2	FLG WN 300# RF LTCS A350 LF2 CL.1 SCH XS	0	0		
1580	U04		FLANGE	5556473	3	3	FLG WN 300# RF LTCS A350 LF2 CL.1 SCH STD	0	0		
1590	U04		FLANGE	9001480	4	4	FLG WN 300# RF LTCS A694 F60 WT 6.02MM	0	0		
1600	U04		FLANGE	9001483	6	10	FLG WN 300# RF LTCS A694 F60 WT 6.35MM	0	0		
1610			FLANGE	9001483	6	10	FLG WN 300# RF LTCS A694 F60 WT 6.35MM	0	0		
1620			FLANGE	9006225	12	12	FLG WN 300# RF LTCS A694 F60 WT 7.09 MM	0	0		
1630			FLANGE	9001455	14	14	FLG WN 300# RF LTCS A694 F60 WT 7.92MM	0	0		
1640	U04		FLANGE	9001457	16	18	FLG WN 300# RF LTCS A694 F60 WT 8.74MM	0	0		
1650	U04		FLANGE	9001458	20	20	FLG WN 300# RF LTCS A694 F60 WT 9.53MM	0	0		
1660	U04		BLIND	5615576	0.75	3	FLG BLIND 300# RF LTCS A350 LF2 CL.1	0	0		
1670			BLIND	9001450	4	20	FLG BLIND 300# RF LTCS A694 F60	0	0		
1680											
1680							***** GASKETS *****				
1690											

	TENGIZCHEVROI TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD SEAL WELDING - REQUIRED PIPE BENDS - INDUCTION BENDS THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 3 MM	SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 47 BARG. @ 90 DEG.C. (LIMITED BY PIPE WT) FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 300 RF TEMPERATURE RANGE: -40 / 90 °C	12/16/2024	BY OZAC	U04	Page 3 of 4	300H5F

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
1690			GASKET	5673033	0.75	20	GASKET 300# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
1700										
1700		B1					***** BOLTING *****			
1710										
1710	U04		BOLTING	5675770	0.5	99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS	0	0	
1720										
1720		SB6	BOLTING	5675675	12	20	JACK SCREW HEX HEAD A320 GR L7M	0	0	
1730										
1730			SPEC BLIND	5749581	0.75	10	SPECTACLE BLIND 300# RF LTCS A516 GR.70	0	0	
1740										
1740			PADDLE	5750100	12	20	PADDLE BLANK 300# RF LTCS A516 GR.70	0	0	
1750			PADDLE	5750200	12	20	PADDLE SPACER 300# RF LTCS A516 GR.70	0	0	
1760										

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD SEAL WELDING - REQUIRED PIPE BENDS - INDUCTION BENDS THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 3 MM	SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 47 BARG. @ 90 DEG.C. (LIMITED BY PIPE WT) FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 300 RF TEMPERATURE RANGE: -40 / 90 °C	12/16/2024	BY OZAC	U04	Page 4 of 4	300H5F